1. OMIM Phenotype Information

Sickle Cell Anemia (HBB gene)

OMIM Entry: 603903 - SICKLE CELL DISEASE

Phenotype Description:
Sickle cell disease is an autosomal recessive disorder characterized by chronic hemolytic anemia, recurrent pain crises, acute chest syndrome, stroke, and progressive multi-organ damage. The phenotype results from polymerization of deoxygenated hemoglobin S (HbS), causing red blood cells to assume a sickle shape, leading to vaso-occlusion and hemolysis. Clinical features include:

System	Manifestations
Hematologic	Chronic hemolytic anemia, reticulocytosis, jaundice, aplastic crises
Vascular	Vaso-occlusive pain crises (dactylitis, bone pain, acute chest syndrome), stroke
Infectious	Increased susceptibility to encapsulated organisms (Streptococcus pneumoniae, Haemophilus influenzae) due to functional asplenia
Organ damage	Splenic infarction (autosplenectomy), renal papillary necrosis, avascular necrosis of femoral head, pulmonary hypertension
Laboratory	HbS on hemoglobin electrophoresis, target cells, Howell-Jolly bodies (asplenia)

Age of onset is typically in early childhood (6-12 months). Heterozygotes (sickle cell trait) are usually asymptomatic and have protection against malaria.

Duchenne Muscular Dystrophy (DMD gene)

OMIM Entry: 310200 - MUSCULAR DYSTROPHY, DUCHENNE TYPE

Phenotype Description:
Duchenne muscular dystrophy is an X-linked recessive disorder characterized by progressive muscle degeneration and weakness due to complete absence of dystrophin protein.

System	Manifestations
Musculoskeletal	Proximal muscle weakness (pelvic girdle first), calf pseudohypertrophy, Gowers' sign (using hands to rise from floor), waddling gait, contractures, scoliosis
Age of onset	Before age 5 years (typically 2-4 years)
Disease progression	Wheelchair-dependent by age 12, respiratory insufficiency by late teens
Cardiac	Dilated cardiomyopathy (by age 18 in nearly all patients)
Respiratory	Progressive respiratory muscle weakness leading to nocturnal hypoventilation, respiratory failure
Laboratory	Markedly elevated serum creatine kinase (CK) (50-100× normal), muscle biopsy shows absent dystrophin staining
Prognosis	Death typically in 20s-30s from respiratory failure or cardiomyopathy

Female carriers may show milder manifestations due to X-inactivation patterns.

Fabry Disease (GLA gene)

OMIM Entry: 301500 - FABRY DISEASE

Phenotype Description:
Fabry disease is an X-linked lysosomal storage disorder caused by deficient alpha-galactosidase A activity, leading to progressive accumulation of globotriaosylceramide (Gb3) in lysosomes throughout the body.

System	Manifestations
Neurologic (PNS)	Acroparesthesias (episodic burning pain in hands/feet, age 4-8 years), pain crises triggered by exercise, fever, stress
Skin	Angiokeratomas (reddish-purple macules/papules, typically in "bathing trunk" distribution), hypohidrosis/anhidrosis
Ophthalmic	Corneal opacities (cornea verticillata - whorl-like pattern), lenticular opacities
Renal	Progressive proteinuria, isosthenuria, renal failure (age 30-50)
Cardiac	Left ventricular hypertrophy, conduction abnormalities, valvular disease, cardiomyopathy
Cerebrovascular	Stroke/TIA (often in young adulthood)
Gastrointestinal	Postprandial abdominal pain, diarrhea, nausea
Laboratory	Deficient alpha-galactosidase A activity (<1% in males), elevated plasma Gb3 and lyso-Gb3
Prognosis	Classic form: death at ~41 years (renal/cardiac/cerebrovascular); late-onset variants: milder symptoms after age 25

Female carriers show variable expression from asymptomatic to severe manifestations due to X-inactivation patterns .

2. ACMG/AMP Classifications

Sickle Cell Anemia (HBB c.20A>T, p.Glu7Val)

ACMG/AMP Classification: Pathogenic

Evidence Code	Criteria	Rationale
PS1	Same amino acid change as previously established pathogenic variant	p.Glu7Val (HbS) is the canonical sickle cell mutation, extensively documented
PS3	Well-validated functional studies	Extensive functional studies demonstrate hemoglobin polymerization, sickling assays, and clinical correlation
PM2	Absent from controls (or extremely low frequency in homozygous state)	Extremely low frequency in homozygous state in population databases (though allele frequency is elevated in some populations due to balancing selection - heterozygous advantage against malaria)
PP4	Phenotype specificity is highly specific for the disorder	Highly specific phenotype (sickle cell disease with characteristic hemolytic anemia and vaso-occlusive crises)

ACMG/AMP Points: This variant meets multiple pathogenic criteria and has been classified as Pathogenic by multiple submitters in ClinVar with no conflicts.

Duchenne Muscular Dystrophy (DMD c.8713C>T, p.Arg2905Ter)

ACMG/AMP Classification: Pathogenic

Evidence Code	Criteria	Rationale
PVS1	Null variant (nonsense) in a gene where LOF is a known mechanism	Nonsense mutation predicted to undergo nonsense-mediated decay, leading to absent protein product

Evidence Code	Criteria	Rationale
PM2	Absent from population databases	Not present in gnomAD or other population databases
PM3	In trans with a pathogenic variant (for recessive disorders)	Observed in hemizygous males (X-linked recessive) with DMD phenotype
PP4	Phenotype specificity highly specific	Highly specific phenotype (Duchenne muscular dystrophy with absent dystrophin staining on muscle biopsy)
PP5	Reputable source reports as pathogenic	Multiple clinical testing laboratories (Invitae, Athena Diagnostics, Eurofins) report as Pathogenic

Supporting Evidence:

- This variant was identified in 6 of 141 patients with Duchenne muscular dystrophy previously found negative for large deletions
- Haplotype analysis indicated all 6 mutations occurred independently
- Loss-of-function variants in DMD are well-established as pathogenic (PMID: 16770791, 25007885)

ACMG/AMP Points: This variant received multiple pathogenic evidence codes and is classified as Pathogenic by all submitters in ClinVar.

Fabry Disease (GLA c.337T>C, p.Cys113Arg)

ACMG/AMP Classification: Pathogenic

Evidence Code	Criteria	Rationale
PS1	Same amino acid change as previously established pathogenic variant	Other pathogenic variants at this residue (p.Cys113Tyr, p.Cys113Phe) are documented
PM1	Located in a mutational hotspot/critical functional domain	Located in alpha-galactosidase A active site region; 42 missense pathogenic changes within ±8 amino acids
PM2	Absent from population databases	Not present in gnomAD or other population databases
PM5	Novel missense change at same residue as another pathogenic variant	Other missense variants at same residue (p.Cys113Tyr, p.Cys113Phe) are pathogenic
PS3	Well-established functional studies	Functional studies demonstrate enzyme deficiency (<1% activity) and Gb3 accumulation
PP3	Multiple lines of computational evidence support deleterious effect	MetaRNN: 0.905 (deleterious), multiple algorithms predict damaging

Supporting Evidence:

- This variant disrupts a critical disulfide bond in the enzyme's active site
- Patient fibroblasts show characteristic lamellar inclusion bodies
- Elevated plasma Gb3 and lyso-Gb3 in affected individuals

ACMG/AMP Points: This variant received 17 ACMG points (17P, 0B) and is classified as Pathogenic by multiple submitters.